

- RAG Learning

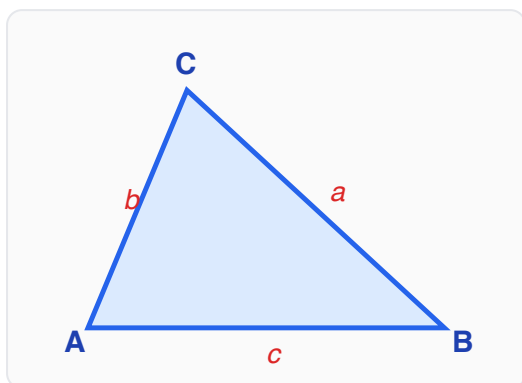
4.1 Angles, Polygons & Constructions

AQA GCSE Mathematics 8300 — Geometry & Measures — spec refs G1–G4, G6, G15 — Papers 1 (non-calculator), 2 and 3 (calculator)

4.1.1 Geometric Terms and Notation

Term	Meaning
Point, line, plane	A point has position only; a line is straight and has length; a plane is a flat surface.
Vertex (plural vertices)	A corner where edges or sides meet.
Edge	A line segment where two faces of a 3D shape meet (or a side of a 2D shape).
Parallel lines	Lines that are always the same distance apart and never meet; marked with matching arrows.
Perpendicular lines	Lines that meet at a right angle (90°), marked with a small square.
Polygon	A closed 2D shape made of straight sides.
Regular polygon	All sides equal and all angles equal.
Line symmetry	The shape can be folded onto itself along a mirror line.
Rotational symmetry	The order is the number of times the shape looks the same in one full turn.

Labelling triangles and angles



- Vertices are labelled with capital letters: triangle ABC.
- The side opposite angle A is labelled with the lower-case letter a , and so on.
- Angle ABC (or $\angle ABC$, or $\hat{A}BC$) is the angle **at B**, between BA and BC. The middle letter is the vertex.
- A line segment between A and B is written AB.

Exam Tip: When a question says “draw a diagram from this description”, sketch it, label every vertex and mark equal sides (dashes), parallel lines (arrows) and right angles (small square). A clear diagram makes the angle facts easier to spot.

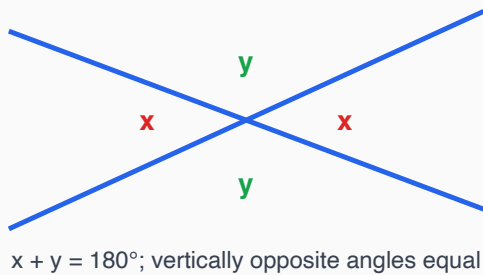
4.1.2 Basic Angle Facts

Type of angle	Size
Acute	Less than 90°
Right angle	Exactly 90°
Obtuse	Between 90° and 180°
Reflex	Between 180° and 360°

Angles on a straight line add up to 180°

Angles around a point add up to 360°

Vertically opposite angles are equal

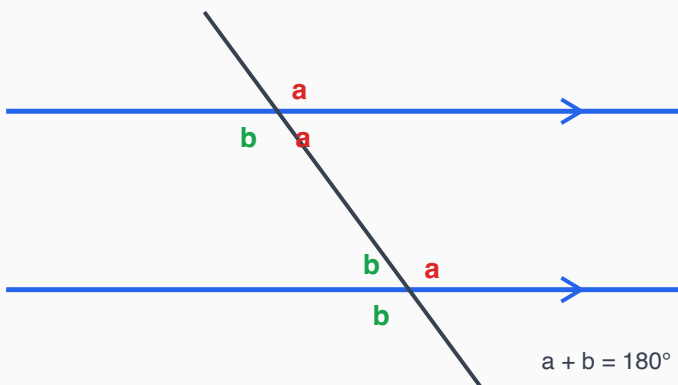


Worked Example: Two angles on a straight line are $2x$ and $3x$. Find x .

$$2x + 3x = 180 \Rightarrow 5x = 180 \Rightarrow x = \mathbf{36^\circ}$$

4.1.3 Angles and Parallel Lines

A straight line that crosses two parallel lines is called a **transversal**. It creates three key pairs of angles.



Alternate angles are equal (on opposite sides of the transversal, between the parallel lines).

Corresponding angles are equal (in matching positions at each parallel line).

Co-interior angles add up to 180° (on the same side of the transversal, between the parallel lines).

Exam Tip: AQA will **not** accept "Z angles", "F angles" or "C angles" as reasons. You must write the full reason, e.g. "alternate angles are equal", "corresponding angles are equal" or "co-interior angles add up to 180° ".

Worked Example: A transversal makes corresponding angles $(3x + 10)^\circ$ and $(5x - 30)^\circ$ with two parallel lines. Find x .

Corresponding angles are equal: $3x + 10 = 5x - 30$

$$40 = 2x \Rightarrow x = 20$$

Worked Example: PQ and RS are parallel, with T a point between them. Angle QPT = 35° and angle SRT = 50° . Find angle PTR.

Draw a line through T parallel to PQ and RS.

The angle above this line at T = 35° (alternate angles are equal).

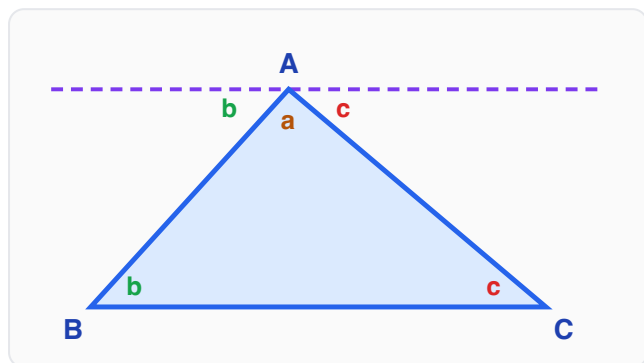
The angle below this line at T = 50° (alternate angles are equal).

$$\text{Angle PTR} = 35 + 50 = 85^\circ$$

4.1.4 Triangles

Triangle	Properties
Equilateral	3 equal sides, 3 angles of 60° , 3 lines of symmetry, rotational symmetry order 3
Isosceles	2 equal sides, 2 equal base angles, 1 line of symmetry
Scalene	No equal sides, no equal angles
Right-angled	One angle of 90°
Acute-angled	All three angles less than 90°

Obtuse-angled

One angle greater than 90° **Proof: angles in a triangle add up to 180°** 

Worked Example: Draw a line through A parallel to BC.

The angle to the left of a equals b (alternate angles are equal).

The angle to the right of a equals c (alternate angles are equal).

These three angles lie on a straight line, so $a + b + c = 180^\circ$. ■

Exterior angle of a triangle

An exterior angle of a triangle equals the **sum of the two interior opposite angles**.

Proof: $a + b + c = 180$ (angles in a triangle) and $c + e = 180$ (angles on a straight line),
so $e = a + b$.

Worked Example: The angles of a triangle are x , $x + 20$ and $2x$. Find x .

$$x + x + 20 + 2x = 180 \Rightarrow 4x = 160 \Rightarrow x = 40^\circ \text{ (angles } 40^\circ, 60^\circ, 80^\circ)$$

4.1.5 Special Quadrilaterals

The angles in any quadrilateral add up to **360°** (it splits into two triangles).

Shape	Sides	Angles	Diagonals	Symmetry (lines / rotation)
Square	4 equal; opposite sides parallel	All 90°	Equal, bisect at 90°	4 / order 4
Rectangle	Opposite sides equal and parallel	All 90°	Equal, bisect each other	2 / order 2
Rhombus	4 equal; opposite sides parallel	Opposite angles equal	Bisect each other at 90°	2 / order 2
Parallelogram	Opposite sides equal and parallel	Opposite angles equal	Bisect each other	0 / order 2
Kite	Two pairs of equal adjacent sides	One pair of opposite angles equal	Cross at 90° ; one bisects the other	1 / order 1
Trapezium	Exactly one pair of parallel sides	Co-interior angles add to 180°	No special property	0 / order 1 (isosceles trapezium: 1 line)

Worked Example: The angles of a quadrilateral are $2x + 10$, $3x$, $4x - 20$ and $x + 30$. Find the largest angle.

$$10x + 20 = 360 \Rightarrow x = 34$$

Angles: 78° , 102° , 116° , 64° . Largest angle = **116°**

Exam Tip: A square is a special rectangle *and* a special rhombus; a rhombus is a special parallelogram. When asked for a property that one shape has but another does not, choose

one that really distinguishes them (e.g. a rhombus has diagonals at right angles, a general parallelogram does not).

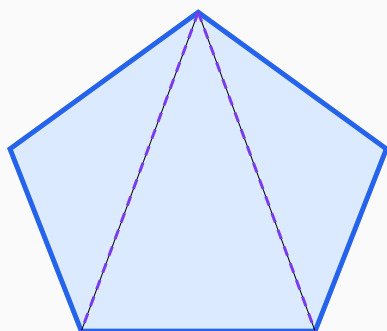
4.1.6 Angles in Polygons

Sides	Name	Interior angle sum	Each interior angle (regular)	Each exterior angle (regular)
3	Triangle	180°	60°	120°
4	Quadrilateral	360°	90°	90°
5	Pentagon	540°	108°	72°
6	Hexagon	720°	120°	60°
8	Octagon	1080°	135°	45°
10	Decagon	1440°	144°	36°

Sum of interior angles of an n -sided polygon $= (n - 2) \times 180^\circ$

Sum of exterior angles of **any** polygon $= 360^\circ$

Regular polygon: exterior angle $= \frac{360^\circ}{n}$, interior angle $= 180^\circ - \text{exterior angle}$



Pentagon = 3 triangles: $3 \times 180^\circ = 540^\circ$

Worked Example: Each interior angle of a regular polygon is 156° . How many sides does it have?

$$\text{Exterior angle} = 180 - 156 = 24^\circ$$

$$n = 360 \div 24 = \mathbf{15} \text{ sides}$$

Worked Example: The interior angles of a polygon add up to 1980° . How many sides does it have?

$$(n - 2) \times 180 = 1980 \Rightarrow n - 2 = 11 \Rightarrow n = \mathbf{13}$$

Worked Example: Explain why a regular polygon cannot have an interior angle of 100° .

Exterior angle = 80° , so $n = 360 \div 80 = 4.5$. The number of sides must be a whole number, so this is impossible.

Exam Tip: Exterior angles are usually the quickest route. To find the number of sides of a regular polygon, find the exterior angle first and divide it into 360° .

4.1.7 Deriving Results and Simple Proofs

You must be able to use angle facts, properties of shapes and congruent triangles to **derive** results and write short proofs, giving a reason for every step.

Base angles of an isosceles triangle are equal

Worked Example: Triangle ABC has $AB = AC$. Draw the bisector of angle A to meet BC at D. $AB = AC$ (given), angle BAD = angle CAD (AD bisects angle A), AD is common. So triangles ABD and ACD are congruent (SAS), and therefore angle ABC = angle ACB. ■

Pythagoras' theorem can also be derived by fitting four congruent right-angled triangles inside a square and comparing areas.

Multi-step angle reasoning Higher

Worked Example: ABC is isosceles with $AB = AC$ and angle $BAC = 36^\circ$. D lies on AC with $BD = BC$. Prove that $AD = BD$.

Angle $ABC = \text{angle } ACB = (180 - 36) \div 2 = 72^\circ$ (base angles of an isosceles triangle)

Triangle BDC is isosceles ($BD = BC$), so angle $BDC = \text{angle } BCD = 72^\circ$

Angle $DBC = 180 - 72 - 72 = 36^\circ$ (angles in a triangle)

Angle $ABD = 72 - 36 = 36^\circ = \text{angle } BAD$

Two equal angles, so triangle ABD is isosceles and $AD = BD$. ■

Worked Example: ABCDEFGH is a regular octagon. Find angle ACD.

Interior angle $= 135^\circ$. Triangle ABC is isosceles ($AB = BC$), so angle $BCA = (180 - 135) \div 2 = 22.5^\circ$.

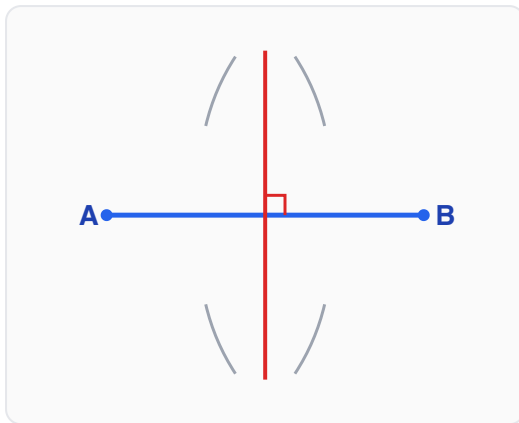
Angle $ACD = 135 - 22.5 = 112.5^\circ$

Exam Tip: In "give reasons" questions, each reason is usually worth a mark. Write the angle fact in words next to each calculation and say *which* triangle is isosceles and why.

4.1.8 Ruler and Compass Constructions

Use a sharp pencil, a ruler and compasses only (no protractor unless told). **Leave all construction arcs showing** — they are the evidence that earns the marks.

Perpendicular bisector of a line segment AB



1. Open the compasses to more than half of AB.
2. With the point on A, draw arcs above and below AB.
3. Keep the same radius; repeat from B so the arcs cross twice.
4. Join the two crossing points. This line cuts AB in half at 90° .

Perpendicular from a point P to a line

1. With centre P, draw an arc that cuts the line at two points, X and Y.
2. Construct the perpendicular bisector of XY (arcs from X and Y on the far side of the line).
3. The line from P through the crossing point is perpendicular to the line.

Perpendicular at a point P on a line

1. With centre P, draw arcs cutting the line at X and Y, equal distances either side of P.
2. Open the compasses wider; draw arcs from X and from Y that cross above P.
3. Join P to the crossing point.

The **shortest distance** from a point to a line is the **perpendicular distance** — measured along the line from the point that meets the given line at 90° .

Bisecting an angle

1. With centre at the vertex O, draw an arc crossing both arms at X and Y.
2. With the same radius, draw arcs from X and from Y that cross at Z.
3. Join OZ. It splits the angle into two equal halves.

Constructing a 60° angle

1. Draw a line and mark a point O on it.
2. With centre O, draw an arc cutting the line at X.

3. With the **same radius** and centre X, draw an arc cutting the first arc at Y.
4. Join OY. Triangle OXY is equilateral, so angle XOY = 60° . Bisect it to get 30° ; bisect a 90° perpendicular to get 45° .

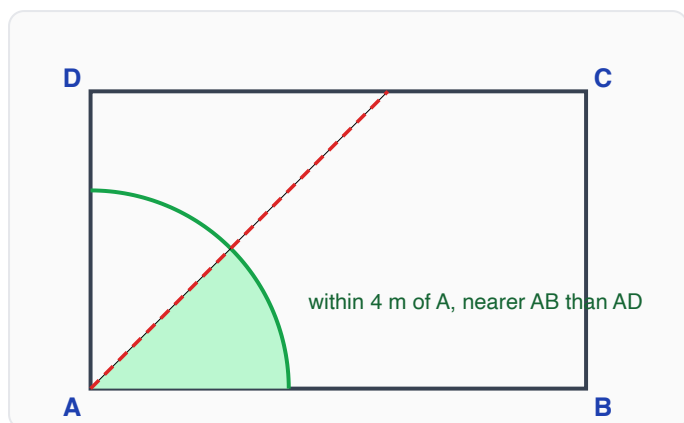
Exam Tip: If you rub out your arcs you will usually lose the marks even if the line is in the right place. Accuracy is normally expected to within 2 mm and 2° .

4.1.9 Loci

A **locus** (plural loci) is the set of all points that obey a given rule.

Rule	Locus
Fixed distance from a point P	A circle centred on P
Fixed distance from a line segment AB	Two parallel lines joined by semicircles at each end (a "running track" shape)
Equidistant from two points A and B	The perpendicular bisector of AB
Equidistant from two intersecting lines	The bisector of the angle between them

"Within 4 m" means the region **inside** the circle; "closer to A than to B" means the side of the perpendicular bisector that contains A. Combine conditions by shading the region that satisfies all of them.



4.1.10 Measuring, Scale Drawings and Bearings

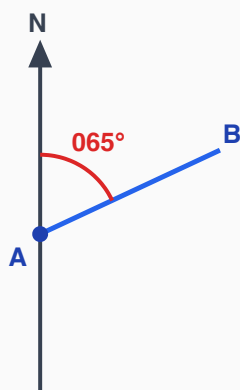
Measuring and scale drawings

- Measure lengths to the nearest millimetre and angles to the nearest degree with a protractor (use the scale that starts from 0° on the arm you are measuring from).
- A **scale** such as 1 cm : 2 km or 1 : 25 000 links lengths on a drawing or map to real lengths.
- Real length = drawing length $\times$ scale factor. Drawing length = real length $\div$ scale factor. Convert units carefully.

Worked Example: A map has scale 1 : 25 000. Two churches are 6.4 cm apart on the map. Find the real distance.

$$6.4 \times 25\,000 = 160\,000 \text{ cm} = 1600 \text{ m} = \mathbf{1.6 \text{ km}}$$

Bearings



A bearing is:

1. measured from **North**
2. measured **clockwise**
3. written with **three figures** (e.g. 065°, 270°)

Back bearing = bearing $\pm 180^\circ$

Compass point	N	NE	E	SE	S	SW	W	NW
Bearing	000°	045°	090°	135°	180°	225°	270°	315°

Worked Example: The bearing of B from A is 065°. Find the bearing of A from B.

$065 + 180 = \mathbf{245}^\circ$ (draw a North line at B: co-interior angles on the parallel North lines add to 180°)

Worked Example: A ship sails from P on a bearing of 050° to A, then on a bearing of 140° to B. Show that angle PAB = 90°.

Bearing of P from A = $050 + 180 = 230^\circ$

Angle PAB = $230 - 140 = \mathbf{90}^\circ$

Exam Tip: "The bearing of B **from** A" means stand at A, face North, and turn clockwise until you face B. Always draw the North line at the point you are measuring *from*.